

Zichen ZHAO

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Research Interests

My research focuses on **Efficient AI and Machine Learning Systems**, with interests in model compression, inference acceleration, large language models, computer vision, and resource-efficient deep learning systems.

Education

Nanjing University of Aeronautics and Astronautics

B.Eng. in Electronic Information and Technology

Overall GPA: 3.4/5.0, Core Coursework GPA: 3.5/5.0

Nanjing, China

Sep 2022 – Jun 2026

Rank: 45 / 159

Publications & Preprints

Paper Title: Weighted Reverse Convolution for Feature Upsampling

Arxiv Preprint

Wentong Li, Zhiyuan Qi, **Zichen Zhao**, Kai Zhang, Lei Zhang

[\[Paper\]](#)

Projects

RECAP: Recovering Compressed LLM Agents with Teacher-guided Soft Prompts 2026.8 – Present

- Developed a soft prompt tuning method to recover the performance of quantized LLMs on agentic tasks via distillation-based training, achieving better effectiveness and efficiency than standard knowledge distillation and SFT while requiring no model weight updates.
- **Keywords:** Agent, Efficient AI, Prompt Tuning.
- **Status:** Ongoing.

Weighted Reverse Convolution for Feature Upsampling

2025.9 – 2026.1

- Developed Weighted Reverse Convolution (WRC), a highly efficient upsampling architecture derived from a Tikhonov-regularized inverse problem formulation. By solving the optimization objective via a differentiable, closed-form FFT solver, the module avoids expensive iterative computations, effectively recovering fine-grained discriminative structures with minimal inference overhead.
- **Keywords:** Computer Vision, Vision Foundation Models, Efficient Feature Upsampling.
- **Status:** Target CVPR 2027.

Improved HandsOnVLM (Temporarily Named)

2025.8 – 2025.9

- Enhanced the Trajectory Decoder by integrating a diffusion-based generative mechanism, enabling more diverse and accurate trajectory predictions. Additionally, investigated its potential to improve multimodal understanding and action reasoning within Vision-Language-Action (VLA) frameworks.
- **Keywords:** Vision-Language Models, Embodied AI.
- **Status:** Exploratory Work (On Hold due to resource reallocation)

Improved YOLOv11 for Aerial Small Object Detection

2025.3 – 2025.8

- Developed an enhanced YOLOv11-based algorithm tailored for UAV aerial imagery. Designed a modified detection head supporting P2–P5 inputs, inspired by the AFPN architecture, and introduced a progressive feature fusion strategy to mitigate semantic gaps and information loss. Achieved moderate improvements on VisDrone2019, raising mAP50 from 34.8% to 38.1% and mAP50–95 from 20.4% to 22.9% over baseline YOLOv11.
- **Keywords:** Computer Vision, Object Detection.
- **Status:** Integrated as a vision module for a UAV-based anti-drone system (the project awarded Grand Prize in the Challenge Cup), providing basic target localization support.

Experience

Westlake University, College of Engineering	Hangzhou, China
Research Intern	2026.4 – Present
• Topic: Efficient AI, Agentic AI. • Project: RECAP: Recovering Compressed LLM Agents with Teacher-guided Soft Prompts. • Supervisor: Huan Wang	
Zhejiang University, College of Computer Science and Engineering	Hangzhou, China
Research Intern	2025.7 – 2025.8
• Project: AI Study Companion	
Nanjing University of Aeronautics and Astronautics, College of AI	Nanjing, China
Undergraduate Research Assistant	2025.7 – 2026.1
• Project: Weighted Reverse Convolution for Feature Upsampling, Improved HandsOnVLM. • Supervisor: Wentong Li	

Skills

Language: CET-6 500.
Programming: Python, C++, Bash
Frameworks: PyTorch, CUDA, Hugging Face
Tools: Git, Docker, Linux, Coding Agents.

Honors & Awards

The Challenge Cup National Competition, Grand Prize	2025
Academic Scholarship, Third and Second Place	2022, 2023, 2024
Outstanding Scholarship, Second Place	2022, 2023, 2024